

3.7 Higher Derivatives

Homework : 1,3,7,15,20,30,35,40,57,59,67

Notation : $\frac{d^n y}{dx^n} = f^{(n)} = D^n f.$

Example 1 : $D^{2007} \cos x = \sin x.$

$$n = 0, \quad \cos x$$

$$n = 1, \quad -\sin x$$

$$n = 2, \quad -\cos x$$

$$n = 3, \quad \sin x$$

$$n = 4, \quad \cos x.$$

Example 2 : $f(t)$ = position function.

$$f'(t) = v(t)$$

$$f''(t) = a(t)$$

$$f'''(t) = j(t) = \text{jerk}.$$

Example 3 : $x^4 + y^4 = 16,$ 求 y'' ?

Solution : $4x^3 + 4y^3 y' = 0 \quad y' = -\frac{x^3}{y^3}$

$$y'' = \frac{-y^3 \cdot (3x^2) + x^3 (3y^2 y')}{y^6}$$

$$= \frac{-48x^2}{y^7}.$$

Example 4 : Let $y = h(x) = f(g(x))$ Find $\frac{d^2 y}{dx^2} = h''(x).$

$$(y = h(x) = f(u), u = g(x), \text{ Find } \frac{d^2 y}{dx^2})$$

Solution : $h'(x) = f'(g(x))g'(x)$

$$h''(x) = f''(g(x))(g'(x))^2 + f'(g(x))g''(x)$$

Example 5 : $y = f(u), u = g(x)$. Find $\frac{d^2y}{dx^2}$

Solution : $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$

$$\Rightarrow \frac{d^2y}{dx^2} = \left(\frac{d}{dx} \left(\frac{dy}{du} \right) \right) \frac{du}{dx} + \frac{dy}{du} \frac{d^2u}{dx^2}$$

$$= \frac{dz}{dx} \frac{du}{dx} + \frac{dy}{du} \frac{d^2u}{dx^2} \quad (\text{Let } z = \frac{dy}{du})$$

$$= \frac{dz}{du} \frac{du}{dx} \frac{du}{dx} + \frac{dy}{du} \frac{d^2u}{dx^2}$$

$$= \frac{d^2y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2u}{dx^2}$$

Example 6 : $f(x) = xg(x^2)$ Find f', f'' .

Solution : $f'(x) = g(x^2) + xg'(x^2)(2x)$

$$= g(x^2) + 2x^2g'(x^2).$$

$$f''(x) = 2xg'(x^2) + 4xg'(x^2) + 4x^3g''(x^2).$$

Remark : Examples 5 and 6 是同一個問題，不同符號的表達方式 .